Q1 We have a nested list representing students and their grades in two subjects:

```
students = [
  ["Alice", [88, 92]],
  ["Bob", [75, 79]],
  ["Charlie", [45, 81]],
  ["David", [60, 65]],
  ["Eva", [90, 95]]
]
```

Tasks:
1.1 Indexing : What would be the result of the following expressions? Fill in the blanks.

students[0][1] →
students[2][1][1] →
students[4][0] →
students[1][1][0] > students[3][1][1] →

1.2 Modifying Nested Lists:

- There has been a grading error and Bob's second score should be increased by 5 points. Write the Python statement that would achieve this.
- David just got a 100 on his third subject! Add this score to his scores.

Output:

1.3 The in operator. What would be the result of the following expressions? Fill in the blanks.

"Alice" in students →
["Alice", [88, 92]] in students →
79 in students[1] →
79 in students[1][1]

1.4

What does the following code output?

```python
def mystery(student):
  num = student[1][0]
  if num > 60:
    if num > 70:
      if num > 80:
        if num > 90:
          return 'A'
        return 'B'
      return 'C'
```

```
        return 'D'
    return 'F'

print(mystery(students[0]))
print(mystery(students[1]))
print(mystery(students[2]))
print(mystery(students[3]))
print(mystery(students[4]))
```

1.5 ???

Q2. In this exercise, we will create a safer divide(a, b) function that checks for valid inputs before performing the division. If the inputs are invalid, the function should return "Invalid input" instead of causing an error. Below is the outline of the function. Your task is to determine the correct Boolean expression to replace ??? in the if statement.

```
def divide(a, b):
    if ???:  # Replace ??? with your final Boolean expression
        return a / b
    else:
        return "Invalid input
```

Let's work on the conditional step by step.

Part 1: Checking the Type of "a"
Before performing division, we need to ensure that "a" is either an integer or a float.

Write a Boolean expression that checks if a is either an int or a float (recall the type function from earlier):

Part 2: Identifying Other Necessary Conditions
Besides checking the type of a, what other conditions must be met for (a/b) to be valid?

Write conditions that b must satisfy to ensure the division operation is safe. You can just describe the conditions in text.

Part 3: Writing Boolean expression for conditions
For each of the above conditions, write the corresponding boolean expression.

Part 4: Combining All Conditions
Now, combine all the conditions from Parts 1 and 3 into a single Boolean expression which can replace the ??? in the function outline.